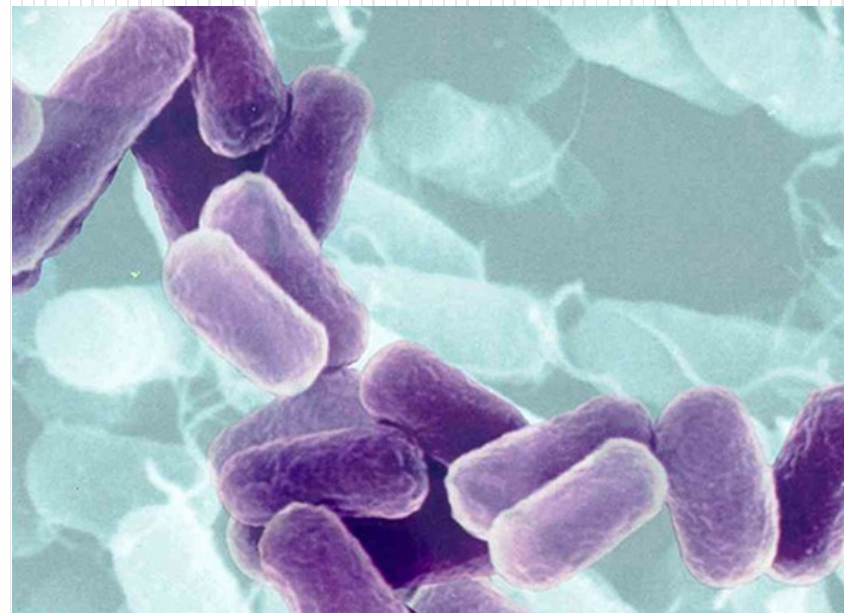


(Phar 3171)



Course Objectives

This course is designed to introduce the students

- ✓ The role of biotechnology and allied technologies in the development of a range of pharmaceutical products of modern biotechnology, and
- ✓ Application of recombinant DNA derived drugs (immunological and biological products) in pharmaceutical care of a patient.

Contents

- ❖ Biotechnology
- ❖ Genetic Engineering
- ❖ rDNA technology
- ❖ Polymerase Chain Reaction (PCR)

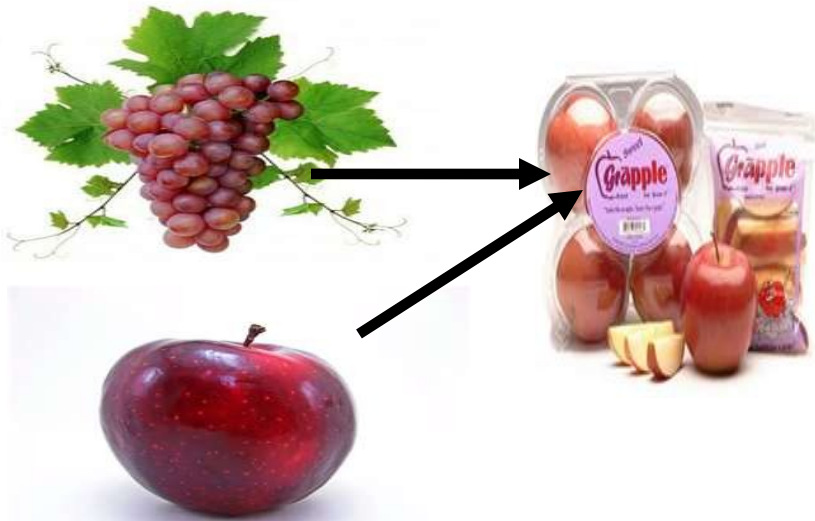
Learning objectives

- **At the end of these topic students should be able to**
 - ✓ **Define biotechnology and pharmaceutical biotechnology.**
 - ✓ **Define genetic engineering.**
 - ✓ **Discuss the concept of rDNA/gene cloning**
 - ✓ **Describe the steps and tools of rDNA technology**
 - ✓ **Discuss the application of rDNA**
 - ✓ **Discribe the concept of PCR technology and its application.**

Introduction.....

- Biotechnology implies the use of microorganisms, plants, and animals or parts thereof for the production of useful compounds.
- Any technique that uses living organisms or substances from those organisms.
 - to make or modify a product,
 - to improve plants or animals, or
 - to develop microorganisms for specific uses

Introduction.....



Introduction

- **Pharmaceutical biotechnology** is biotechnological manufacturing of pharmaceutical products.
- It is the application of genetic engineering and DNA technology to produce therapeutic and medical diagnostic products.
- In recent years, biotechnology drug products have gained a major share of prescribed pharmaceuticals.

Introduction....

- These drug products include
 - vaccines
 - peptides and proteins, including monoclonal antibodies
 - antisense oligonucleotides
 - Therapeutic hormones, growth factors
 - DNA preparations for gene therapy.
 - Therapeutic enzymes

Genetic Engineering(GE).....

- Is artificially manipulation, modification and combination of DNA to modify the genetic makeup of an organism.
- The simple addition, deletion, or manipulation of a single trait in an organism to create a desired change.
- It allows genes from one organism to be inserted into a cell of a different organism of a different species.
- Human genes can be inserted into bacteria and other animals

Genetic Engineering.....

GE process involves:

- Identifying genes
- Isolating genes
- Modifying genes
- Re-expressing genes in other hosts or organisms

Genetic Engineering.....

GE allows:

- Identifying new genes and the proteins they encode
- To correct endogenous genetic defects
- To manufacture large quantities of specific gene products such as hormones, vaccines, and other biological agents of medical interest

Recombinant DNA(rDNA) technology/Gene cloning

- It takes genetic material from one source and recombining it in vitro with another source followed by introducing recombined material into host cell.
- is the science of cutting and recombining DNA from different species to create a molecule that does not exist in nature(rDNA).
- **rDNA** is a recombinant molecule where the **vector is joined with a natural or a synthetic DNA segment of interest** to make a molecule that can replicate in a living cell.
- **vector** : Self-replicating DNA molecules used to transfer foreign DNA segments between host cells.

Steps involved in rDNA technology

1. Isolation
2. Cutting
3. Ligation into cloning vector
4. Transformation
5. Expression

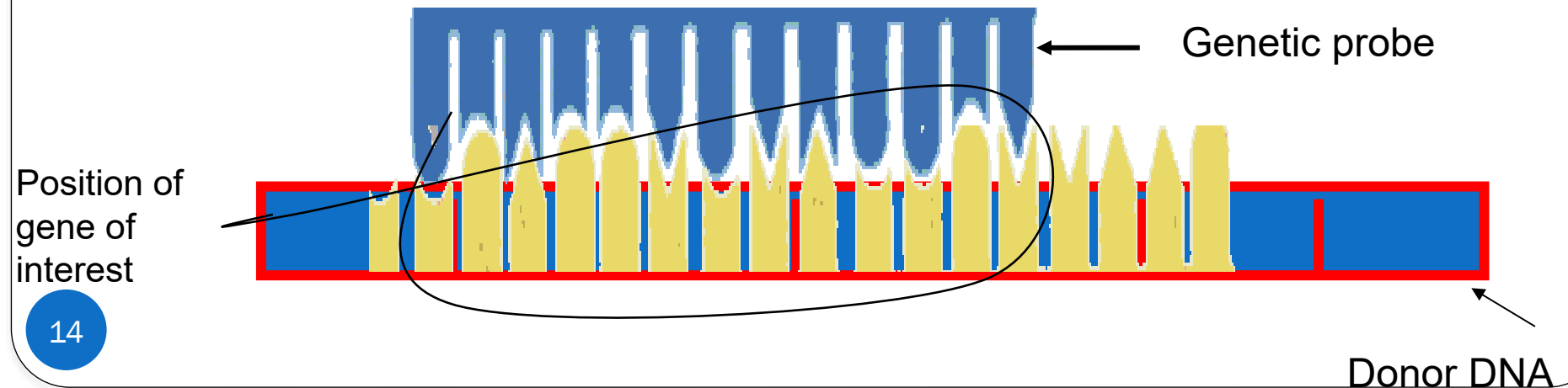
1. Isolation

(a) **Isolation** of a specific gene from donor e.g. human

Can be done by:

- disrupting lipid membranes with detergents
- destroying proteins with phenol or protease
- degrading RNAs with RNase leaving DNA at the end

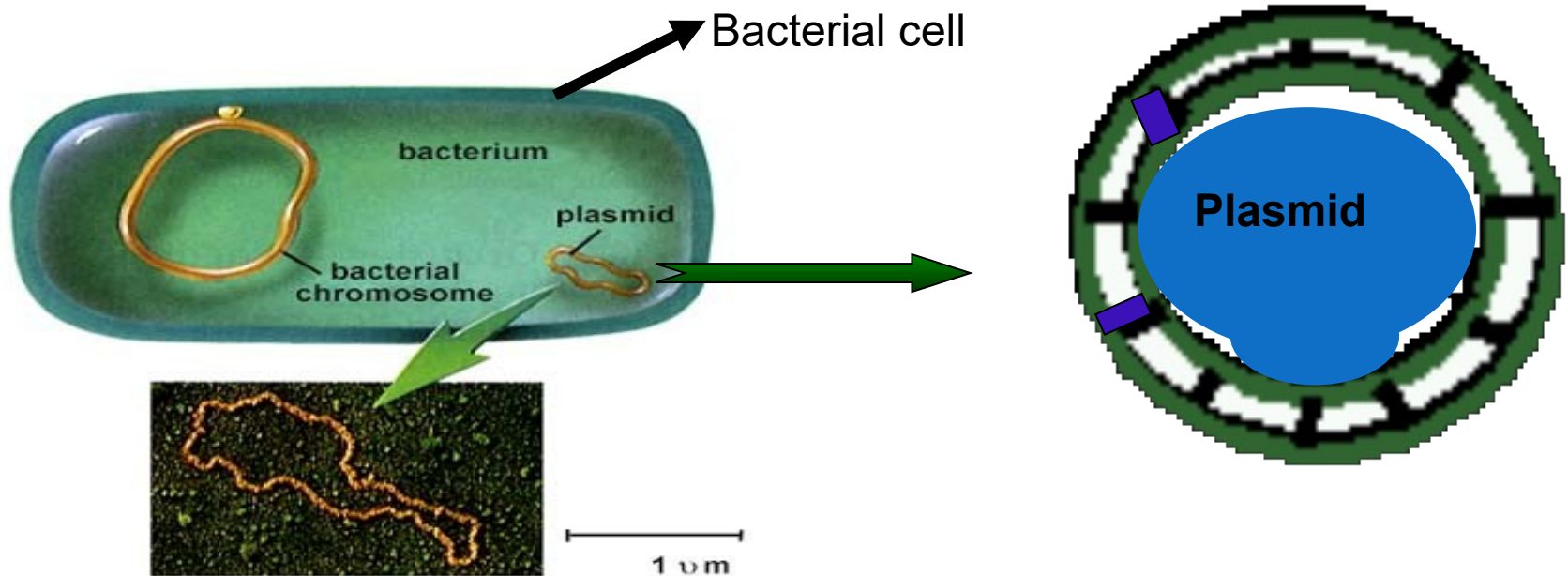
- Genetic probe reveals the position of the gene of interest



1. Isolation

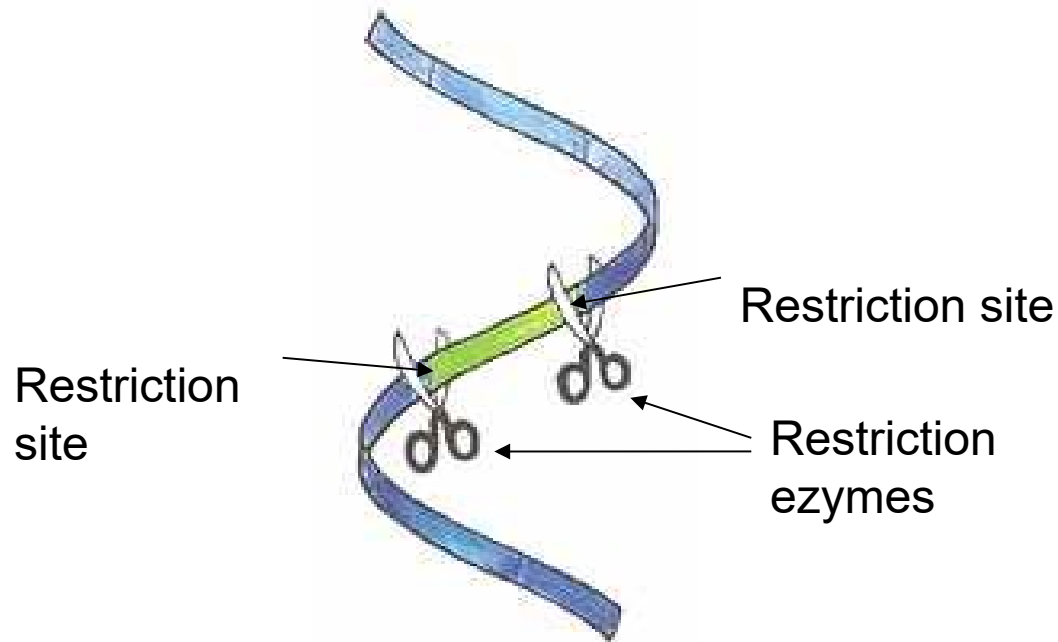
(b) **Isolation** of plasmid from a bacterial cell can be accomplished by

- an alkaline lysis procedure or
- by boiling cells which removes bacterial chromosomal DNA from plasmid DNA.

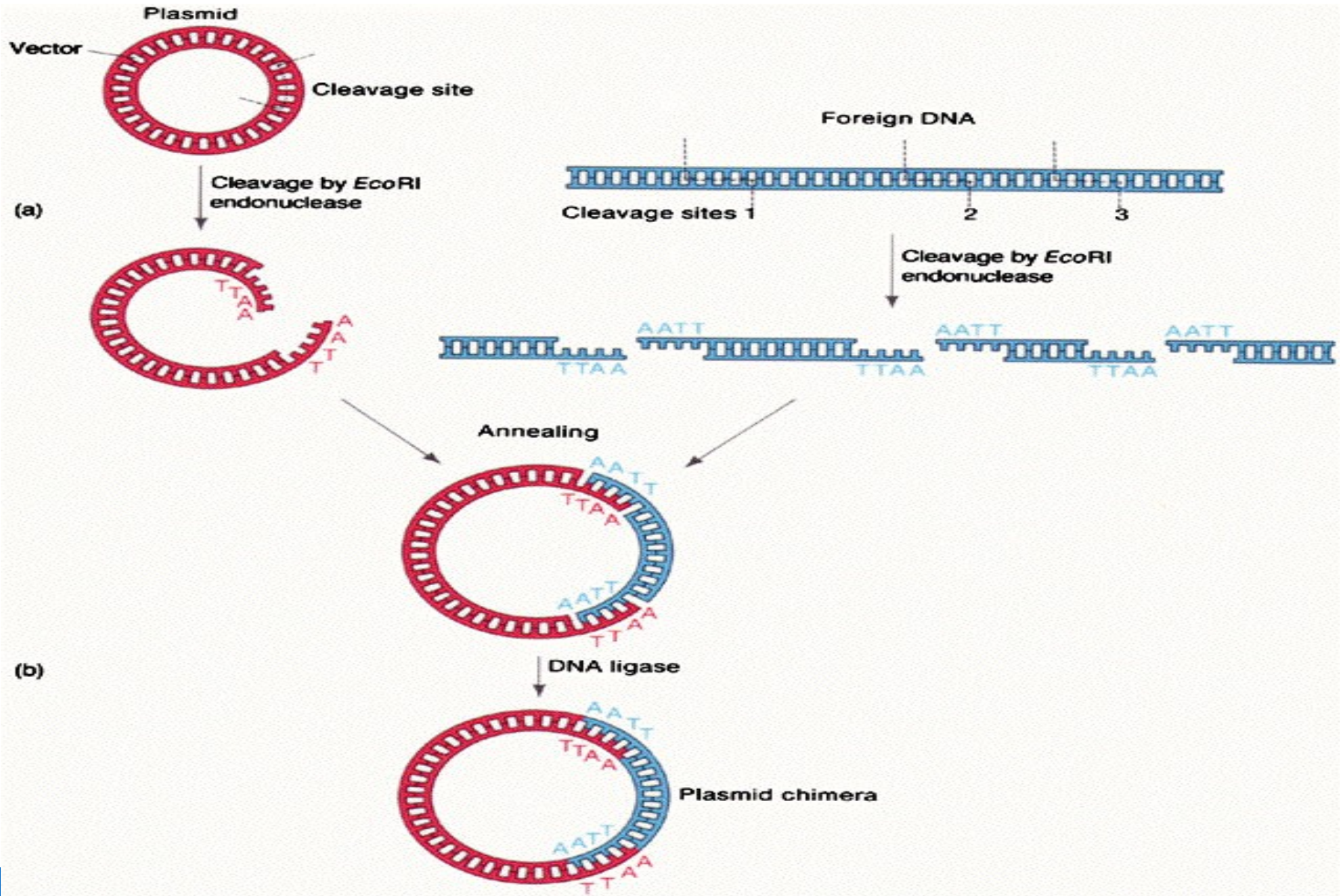


2. Cutting

- ✓ **Restriction enzymes(RE)** are endonucleases that will recognize specific nucleotide sequences in the DNA and break the DNA chain at those points.
- ✓ are the scissors of molecular genetics.



2. Cutting....



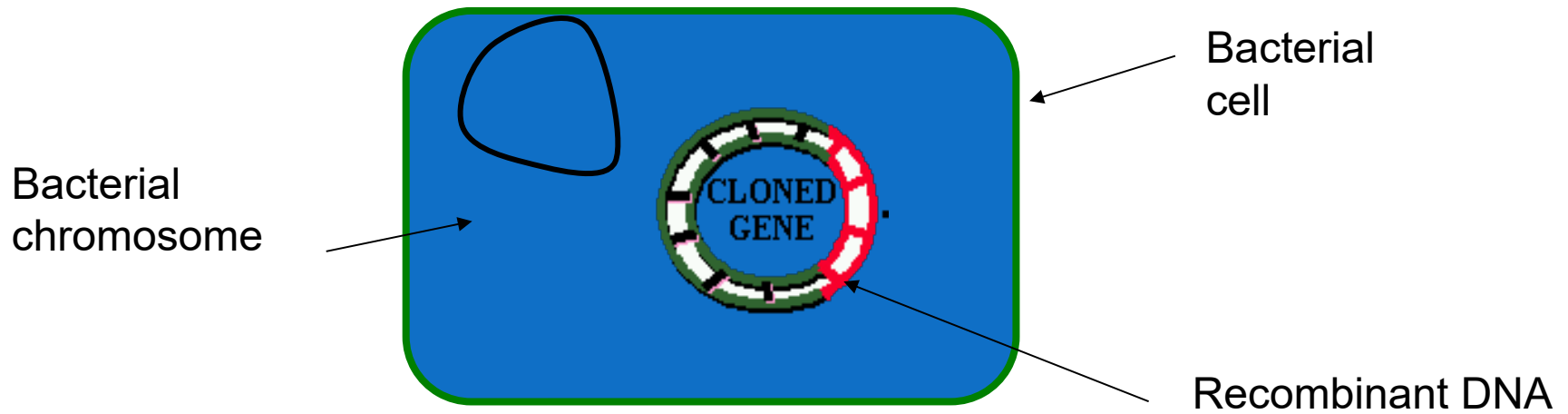
3. Ligation ..

- Once you have isolated and cut the donor and vector DNAs, they must be joined together. The DNAs are mixed together in a tube.
- If both have been cut with the same RE, the ends will match up because they are sticky.
- **DNA ligase** is the glue of molecular genetics that holds the ends of the DNAs together.
- DNA ligase creates a phosphodiester bond between two DNA ends.

4. Transformation

Recombinant DNA introduced into bacterial cell(host cell)

- Can be done by
 - ✓ non-physiological conditions are created by applying a heat shock to the host cells in the presence of high amounts of Ca^{2+} ions.
 - ✓ Electroporation

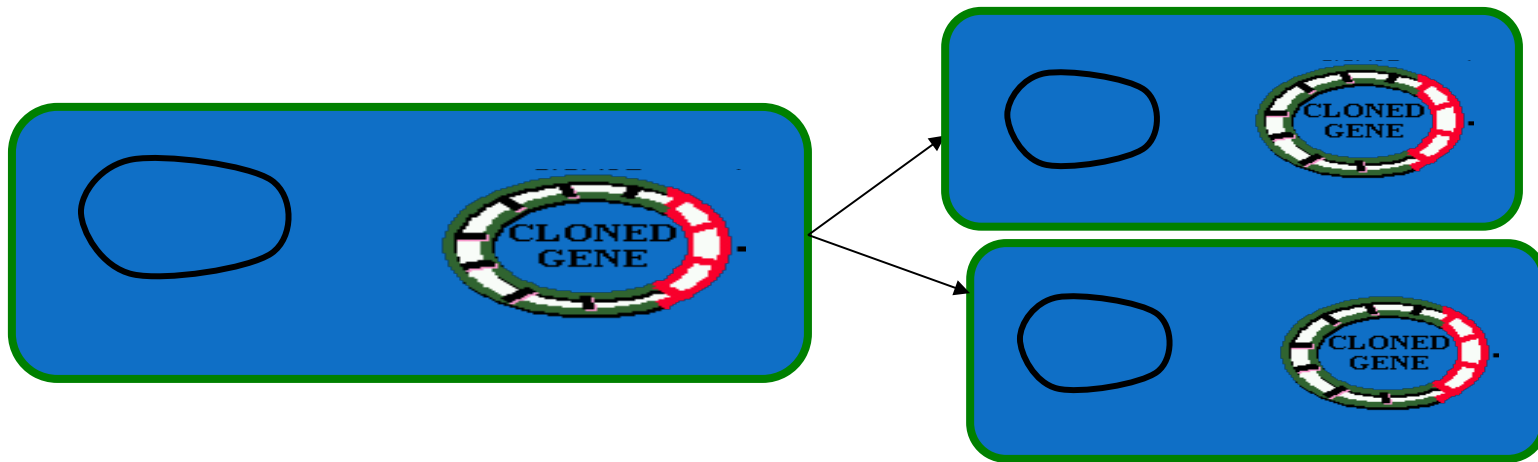


5. Expression

Expression is getting the organism with the recombinant DNA to produce the desired protein

Bacterial cell reproduces by Binary Fission

Bacterial cell produces the polypeptide



Expression.....

- Proteins that can result from the expression of recombinant DNA within living cells **are termed recombinant proteins.**
- When the protein is produced in large amounts it is **isolated and purified.**

Tools of genetic engineering/rDNA

- rDNA requires the following tools
 - ✓ Vectors
 - ✓ Enzymes
 - ✓ Cloning hosts

Vectors

A DNA molecule, capable of replicating in a host organism, into which a gene is inserted to construct a recombinant DNA molecule

Requirements for a cloning vector

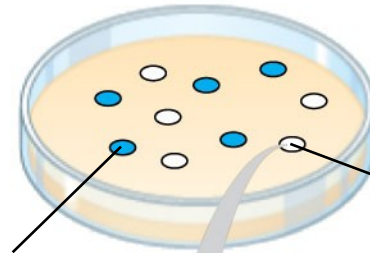
- They must be **autonomously replicating** DNA molecules in the host cell.
- Should have convenient RE sites for inserting DNA of interest
- Should be small and easy to isolate
- Should have a **selectable marker** to indicate which host cells received recombinant DNA molecule and which do not .

Vectors /selectable marker/B galactosidase expreesion/

Bacteria carrying
plasmids



RESULTS



Colony carrying non-
recombinant plasmid
with intact *lacZ* gene

Colony carrying
recombinant
plasmid
with disrupted
lacZ gene



One of many
bacterial
clones

- If B- galactosidase is produced, galactose is hydrolyzed to form 5 bromo 4 chloro-indoxyl which spontaneously dimerizes to produce an **insoluble blue pigment**.

Types of vector

Bacterial plasmids

- are small, circular DNA molecules that are separated from the rest of the chromosome.
- They replicate independently of the bacterial chromosome.
- Useful for cloning DNA inserts less than 20 kb (kilobase pairs).
- Inserts larger than 20 kb are lost easily in the bacterial cell.

Vector.....

Bacteriophage lambda

Bacteriophages or simply phages

- are the viruses that replicate within the bacteria.
- contains a central region of 15 kb that is not required for replication or formation of progeny phage in *E. coli*.
- Thus, lambda can be used as a cloning vector by replacing the central 15 kb with 10-15 kb of foreign DNA.

vector..

Cosmids

- Cosmids combine essential elements of a plasmid and Lambda systems.
- They can replicate like plasmids but can be packaged like phage lambda.
- Cosmids are extracted from bacteria and mixed with restriction endonucleases.

vector..

Cosmids....

- Cleaved cosmids are mixed with foreign DNA that has been cleaved with the same endonuclease.
- Recombinant cosmids are packaged into lambda capsids
- Recombinant cosmid is injected into the bacterial cell where the rcosmid arranges into a circle and replicates as a plasmid.

Enzymes

- The key enzymes used in the recombinant DNA technology are:
 - Nucleases
 - Ligases
 - Polymerases

Enzymes

- **Nucleases**

are the enzymes which degrade the nucleic acid (here DNA). They act on the DNA by breaking the phosphodiester bond.

There are two major types of Nuclease enzyme, namely, Exonuclease and Endonuclease

Exonucleases,

removes the nucleotides one at a time from the end of the DNA molecule.

Endonucleases aid break internal phosphodiester bonds within a DNA molecule.

Enzymes

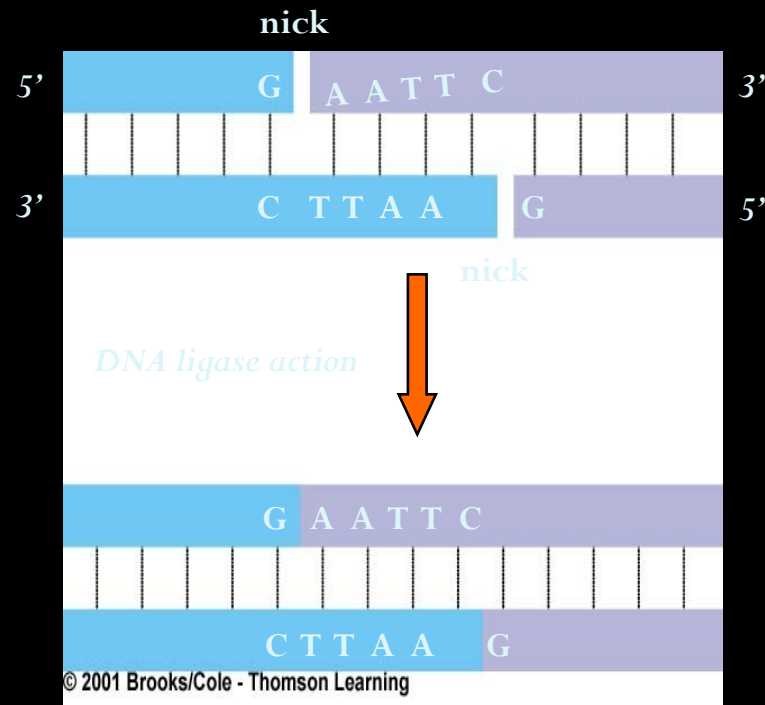
- The restriction endonuclease is a class of enzyme produced by the bacteria which identifies the particular sequence and cleaves explicitly.
- Example of restriction endonuclease is EcoRI, BamHI(bacillus amyoli), PvuI(proteus hauseri), etc

Enzymes....

Ligases

- The DNA ligases have the function of sealing the DNA nicks.
- In the cell, if the phosphodiester bond between two nucleotides of the DNA molecule is not formed or damaged, then DNA ligases are recruited to repair.
- These enzymes aid in the formation of the phosphodiester bond between the two nucleotides.
- In rDNA technology, ligases catalyse the joining of DNA of interest called as 'insert', with the vector molecule and the reaction is **known as ligation**.

Ligases....



Enzymes...

Polymerases

- DNA polymerases are enzymes that catalyze the synthesis of a new DNA strand from a pre-existing strand.
- The enzyme adds deoxyribonucleotides to the free 3'-OH of the chain undergoing elongation.

A—T—G

T—A—C—T—G—G—T—A—G

A—T—G—A—C—C—A—T—C

T—A—C—T—G—G—T—A—G

Enzymes.....

- **It has three major requirements for its activity;**
 - ❖ a template strand for which the enzyme synthesizes a complementary strand;
 - ❖ a primer with a free 3'-OH group that hybridizes with the template to form a double stranded region that initiates the polymerization and
 - ❖ a pool of all the four dNTPs that are used to synthesize the new DNA strand.
 - ❖ In addition, some cofactors like Mg^{2+} ions may be required in a buffer solution with correct pH for optimum activity.

Expression host

- The organism in which the recombinant DNA is to be inserted is called **cloning host or expression host**
- Prokaryotic (bacteria) or eukaryotic (yeast, mammalian cell culture) systems are generally used as a host for the production of usable quantities of the desired rDNA product.
- Most of the rDNA products approved by FDA are being produced using these systems.

Expression host...

Bacteria such as *Escherichia coli*

- are widely used for the expression of rDNA products.
- They offer several advantages due to high level of recombinant protein expression, rapid growth of cell and simple media requirement.
- However, there are some limitations such as intracellular accumulation of heterologous proteins, the potential for product degradation due to trace of protease impurities and production of endotoxin.

Expression host...

- Yeast such as *Saccharomyces cerevisiae*, *Hansenula polymorpha* and *Pichia pastoris* are among the simplest eukaryotic organisms.
- They grow relatively quickly and are highly adaptable to large-scale production.
- These organisms do not produce endotoxin.
- They are capable of glycosylating proteins up to a certain extent like mammalian cells.

Expression host...

- Mammalian systems such as Chinese hamster ovary (CHO) cell and baby hamster kidney (BHK) cell systems are often the choice for production of human therapeutic proteins.
- The CHO and BHK cell systems are an ideal choice as these are capable of glycosylating the protein at the correct sites.
- However, cost of production of the products using these systems is high because of slow growth and expensive nutrient media.

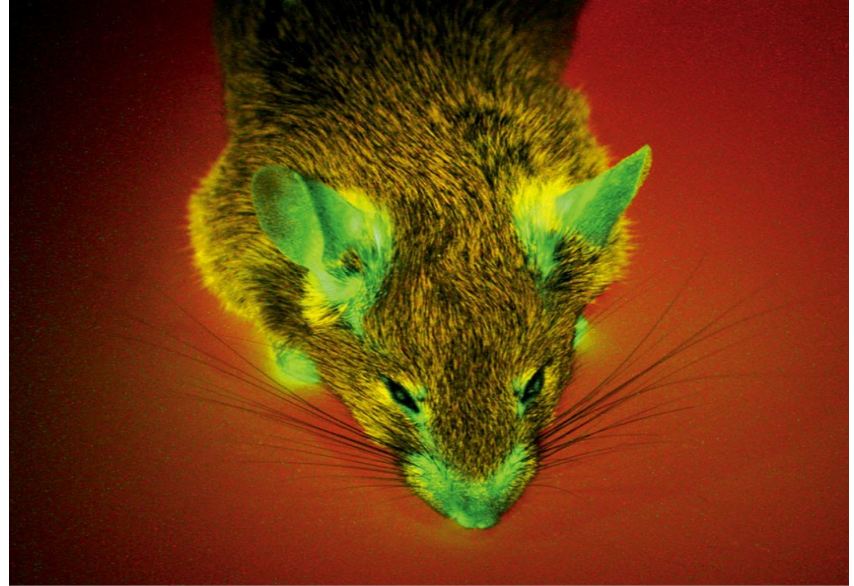
Expression host...

Transgenic Animals

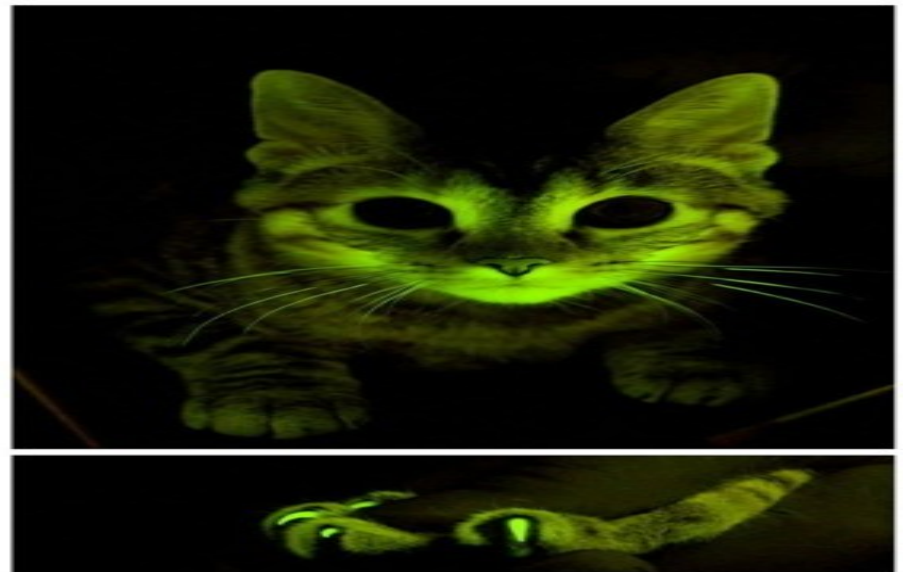
- After recombinant DNA is inserted the organisms will express the new gene and the organisms are genetically modified.
- Genetically Modified organism are called **transgenic organism**.
- **Human genes are now routinely transferred into animals to produce human proteins for use as drugs.**
- Transgenic animals are pharmaceutical “factories,” producers of large amounts of otherwise rare substances for medical use.

Transgenic Mouse

- With genes for fluorescent protein



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Expression host...

Transgenic Plants

- Contain DNA from another species
- New genes make crop plants less vulnerable to disease and pests

Expression host...

Transgenic Cotton

- Resistant to herbicides
- Insecticide gene



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Applications of genetic engineering/r DNA....

- One benefit of DNA technology is identification of human genes in which mutation plays a role in genetic diseases
- **Gene therapy** is the alteration of an afflicted individual's genes
- **Transgenic animals are pharmaceutical “factories,” producers of large amounts of rare substances for medical use.**
- **Recombinant proteins are extensively used as reagents in laboratory experiments and to make antibody probes for investigative protein synthesis within cells and organisms.**

Applications of genetic engineering r DNA

- Several proteins are created from recombinant DNA (recombinant proteins) and are used in **medical applications, particularly as medications and vaccines.**
- On the other hand, **various proteins can also be got from biological sources** made from animal extracts or man-made.

Applications of r DNA....

Recombinant protein used in medical practice care

- **Insulin:** for the treatment of type 1 diabetes
- **Human recombinant growth hormone:** For promoting growth in an individual.
- **Erythropoietin alpha:** Treatment of anemia associated with renal failure, HIV.infection, cancer
- **Coagulation factor VII used for treatment of hemophilia A**

Applications of r DNA....

- **Coagulation factor IX:** Treatment of hemophilia B
- **Dornase alpha:** Treatment of cystic fibrosis.
- **Anti coagulant(Lepirudin):** Prevention of blood clot.
- **Alteplase :** Treatment of acute myocardial infarction
- **Interferon B :** For treatment of multiple sclerosis.
- **Interleukin-2:** Renal cell carcinoma

Polymerase Chain Reaction/PCR

- A method for **amplifying DNA segments of interest** by **in vitro DNA synthesis**, occurring in a number of fast repeating steps.
- PCR copies a DNA molecule without restriction enzymes, vectors or host cells.
- Faster and easier than conventional cloning.

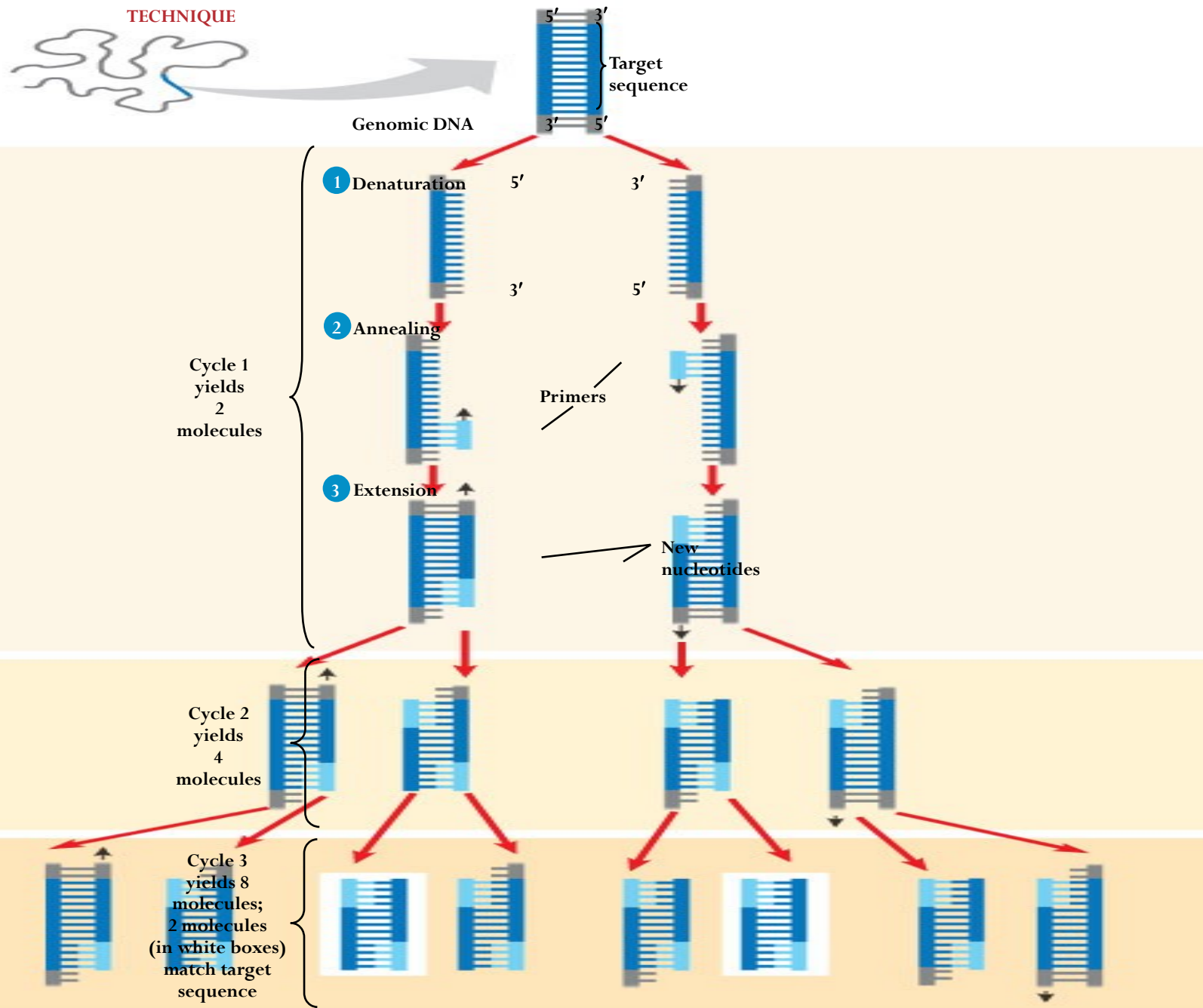
PCR.....

Steps in PCR

1. Denaturation:

- This is done by heating the sample to 92 °C to break the hydrogen bonds between the two polynucleotide strands.
- Two single-stranded DNA molecules serve as templates.

TECHNIQUE



PCR....

2. Annealing:

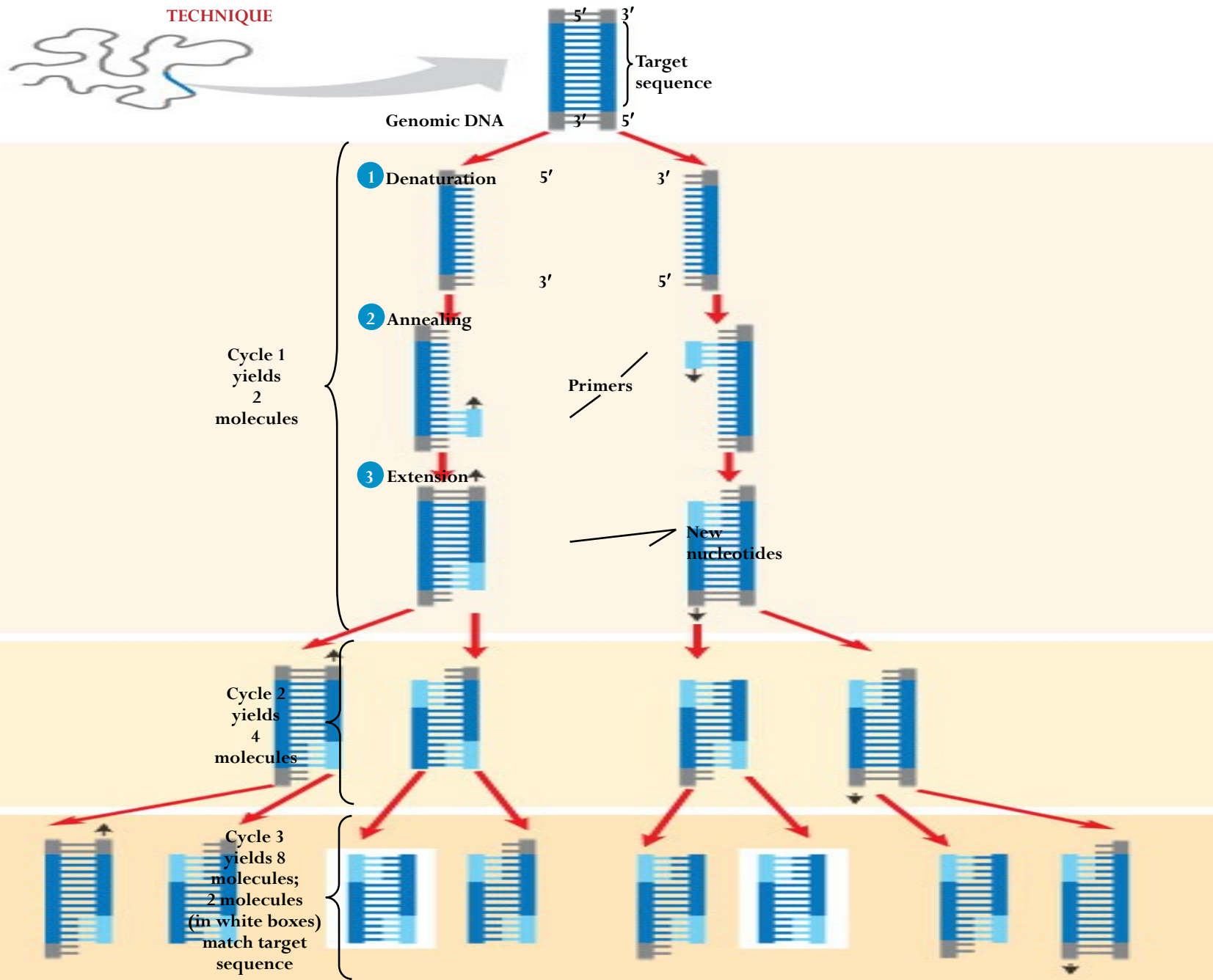
- Short nucleotide sequence (primers for DNA replication) are mixed with the DNA and bind to complementary regions on single-stranded DNA.
- The primers used should be specific complementary to the target region.
- are 20-30 nucleotides long, synthesized in the laboratory.
- The optimal temperature for this process is about 55°C

PCR.....

3.DNA Synthesis:

- The enzymes Taq polymerase is added to synthesize a complementary DNA strand.
- The optimal extension temperature is about 72°C .
- Taq is a DNA polymerase from a bacterium found in hot springs.
- These three steps make up one PCR cycle.

TECHNIQUE



Application of PCR

- PCR allows identification/isolation of DNA fragments from genomic DNA by selective amplification of a specific region of DNA.
- It is used in the detection of a target DNA from minute amount of samples where a substantial amount of DNA is required to allow the detection of target DNA.
 - **the presence of microbial pathogens in raw and processed food products**
 - **Infectious diseases(bacteria and viral)**
 - **Forensic science(genetic fingerprinting)**
 - **Paternity testing**



???